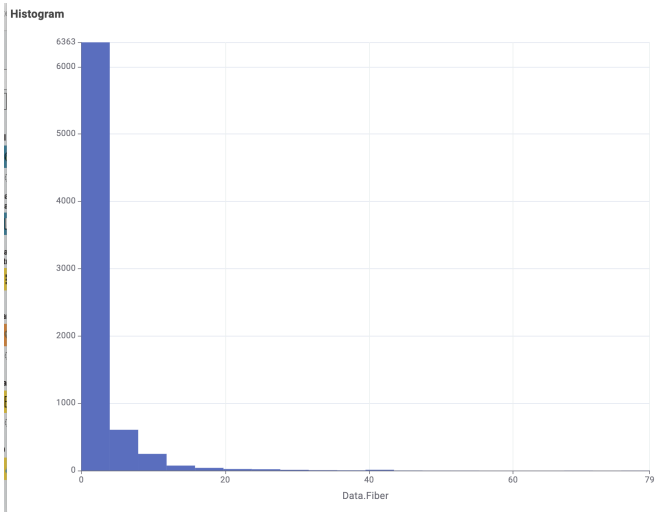
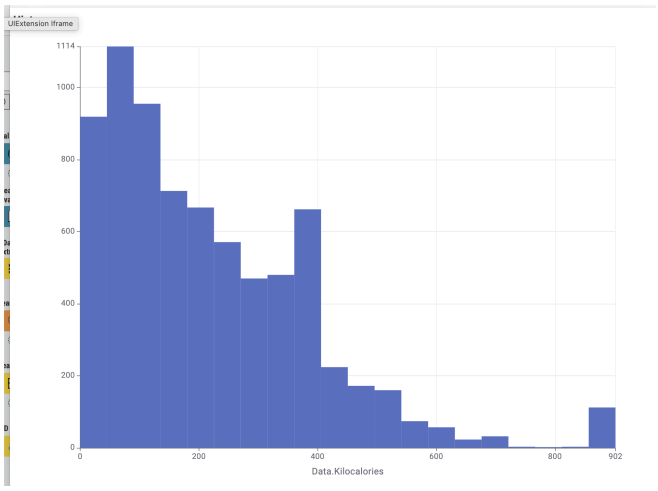
Data Understanding & Profiling

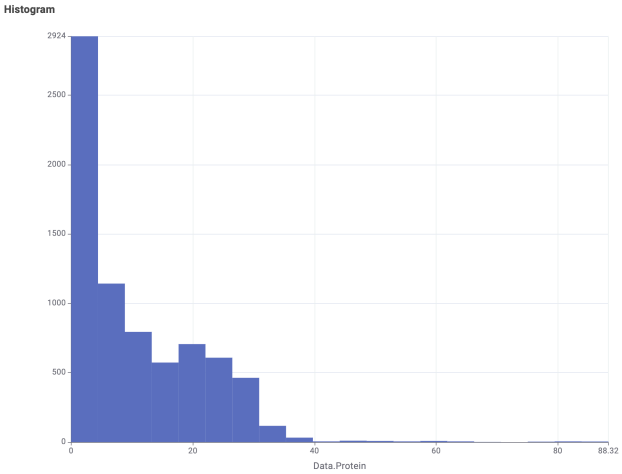
Deliverable 2: Data Audit Dashboard & Team Discussion Form

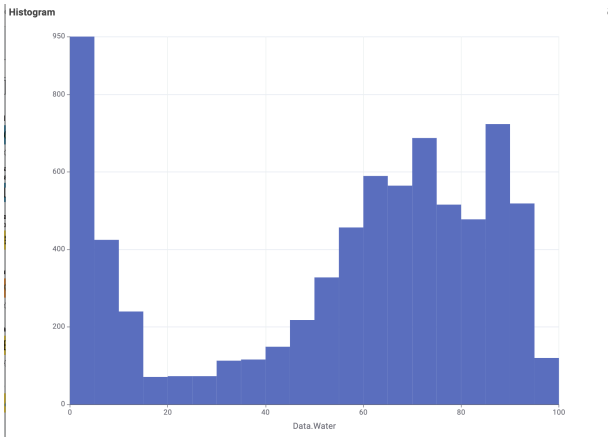
- **Distribution Audit:** Generate histograms and box plots for nutritional features.

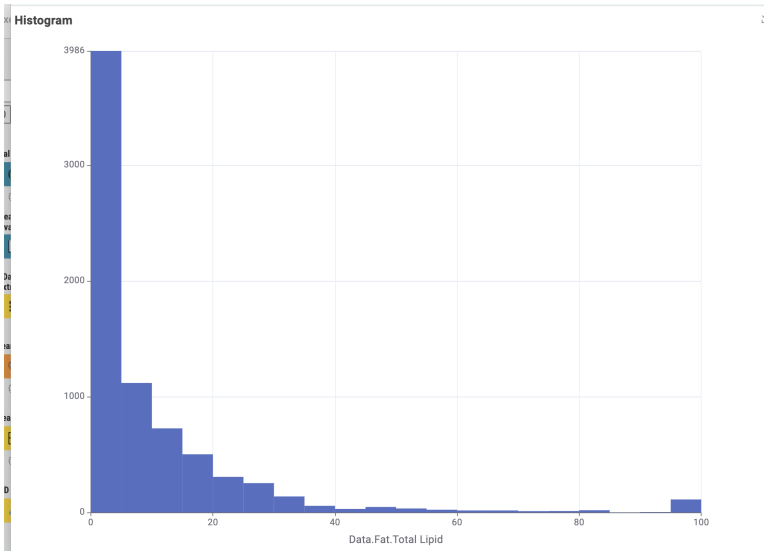


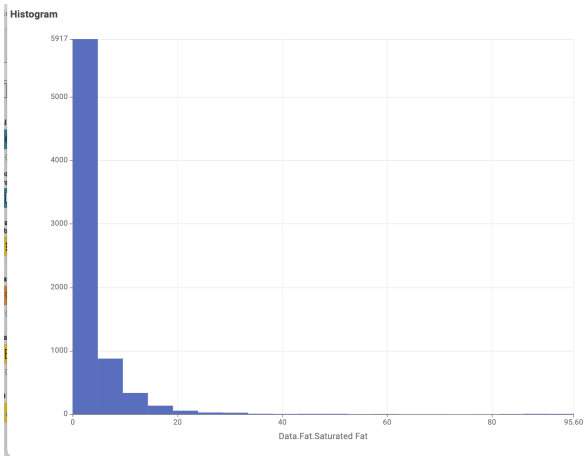


C

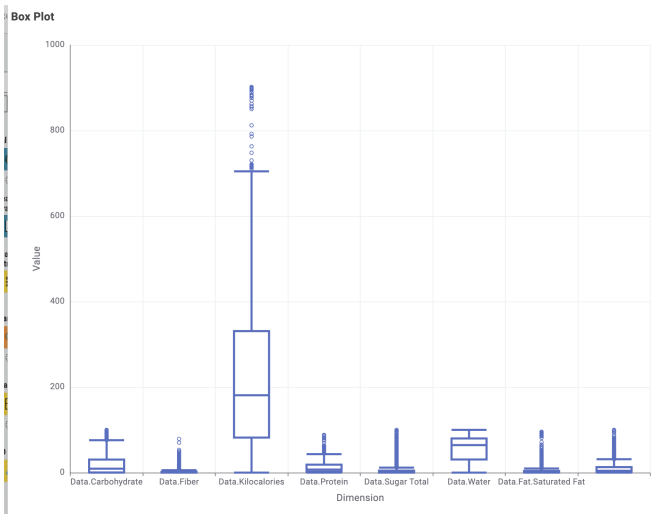

Histogram
2924
2500
2000
1500
1000
500
0
0
20
40
60
80
88.32
Data.Protein


Histogram
950
800
600
400
200
0
0
20
40
60
80
100
Data.Water


Histogram
3986
3000
2000
1000
0
0
20
40
60
80
100
Data.Fat.Total Lipid

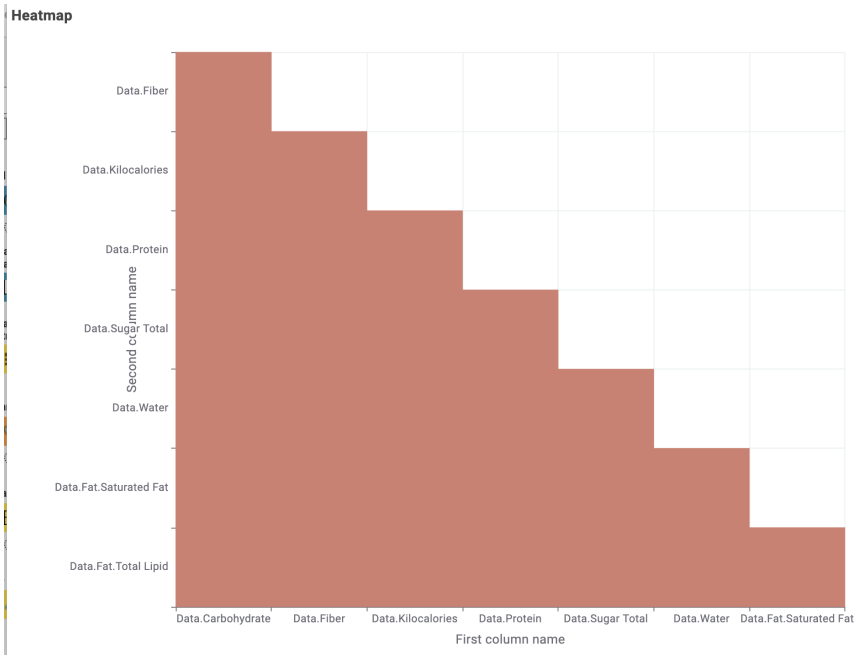


The histogram shows the distribution of kilocalories across the grocery items. Most products fall within the lower calorie ranges, while a smaller number of items have very high calorie values. This suggests that the dataset contains many lower-calorie foods with a few high-calorie outliers.



The box plot visualizes the distribution and spread of the nutritional features in the dataset. Several nutrients show outliers, particularly in fat and calories, indicating that some foods contain extremely high values compared to the rest. The median and quartile ranges help illustrate how nutrients vary across the inventory.

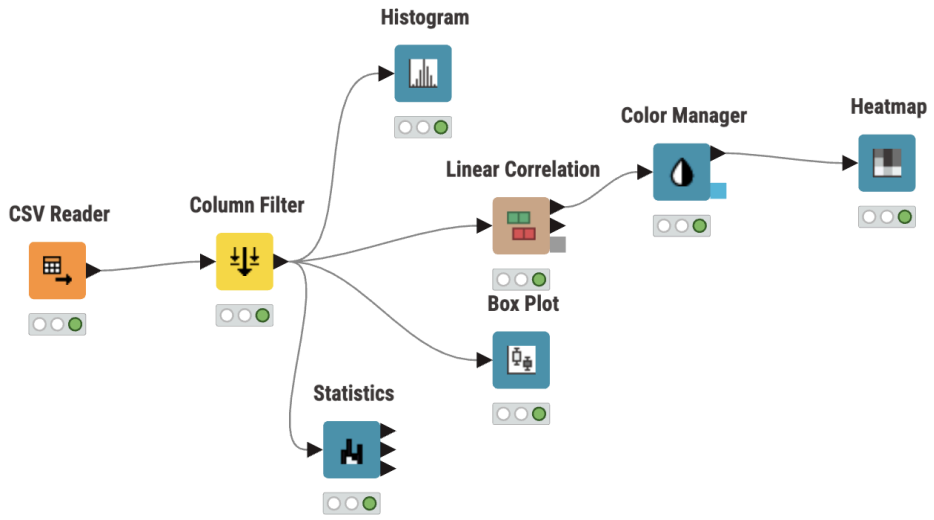
- **Correlation Matrix:** Identify which nutrients tend to appear together (e.g., high sugar and low fiber).



Rows: 28 | Columns: 5

#	RowID	First column name (String)	Second column name (String)	Correlation value (Number (Float))	p value (Number (Float))	Degrees of freedom (Number (Integer))
1	Row0	Data.Carbohydrate	Data.Fiber	0.431	0	7411
2	Row1	Data.Carbohydrate	Data.Kilocalories	0.453	0	7411
3	Row2	Data.Carbohydrate	Data.Protein	-0.252	0	7411
4	Row3	Data.Carbohydrate	Data.Sugar Total	0.601	0	7411
5	Row4	Data.Carbohydrate	Data.Water	-0.757	0	7411
6	Row5	Data.Carbohydrate	Data.Fat.Saturated Fat	-0.085	0	7411
7	Row6	Data.Carbohydrate	Data.Fat.Total Lipid	-0.094	0	7411
8	Row7	Data.Fiber	Data.Kilocalories	0.179	0	7411
9	Row8	Data.Fiber	Data.Protein	-0.034	0.003	7411
10	Row9	Data.Fiber	Data.Sugar Total	0.096	0	7411
11	Row10	Data.Fiber	Data.Water	-0.368	0	7411
12	Row11	Data.Fiber	Data.Fat.Saturated Fat	-0.077	0	7411
13	Row12	Data.Fiber	Data.Fat.Total Lipid	-0.03	0.01	7411
14	Row13	Data.Kilocalories	Data.Protein	0.157	0	7411
15	Row14	Data.Kilocalories	Data.Sugar Total	0.281	0	7411
16	Row15	Data.Kilocalories	Data.Water	-0.891	0	7411
17	Row16	Data.Kilocalories	Data.Fat.Saturated Fat	0.613	0	7411
18	Row17	Data.Kilocalories	Data.Fat.Total Lipid	0.806	0	7411
19	Row18	Data.Protein	Data.Sugar Total	-0.23	0	7411
20	Row19	Data.Protein	Data.Water	-0.161	0	7411
21	Row20	Data.Protein	Data.Fat.Saturated Fat	0.064	0	7411
22	Row21	Data.Protein	Data.Fat.Total Lipid	0.072	0	7411
23	Row22	Data.Sugar Total	Data.Water	-0.435	0	7411
24	Row23	Data.Sugar Total	Data.Fat.Saturated Fat	0.027	0.022	7411
25	Row24	Data.Sugar Total	Data.Fat.Total Lipid	-0.033	0.004	7411
26	Row25	Data.Water	Data.Fat.Saturated Fat	-0.351	0	7411
27	Row26	Data.Water	Data.Fat.Total Lipid	-0.473	0	7411
28	Row27	Data.Fat.Saturated Fat	Data.Fat.Total Lipid	0.762	0	7411

The correlation heatmap highlights relationships between nutritional variables. A strong positive correlation can be observed between total fat and kilocalories, suggesting that foods with higher fat content generally have higher calorie values. Sugar also shows a positive relationship with carbohydrates since sugar is part of total carbohydrates. Additionally, water content tends to have a negative correlation with calories, meaning foods with higher water content are typically lower in calories.



This KNIME workflow performs the exploratory data analysis for the grocery nutrition dataset. The pipeline loads the dataset using the CSV Reader, removes non-numeric columns with the Column Filter, and generates summary statistics. Visualization nodes such as Histogram and Box Plot are used to analyze distributions and identify outliers. Linear Correlation and Heatmap are used to examine relationships between the nutritional variables.

Conclusion:

Overall, the exploratory analysis helps identify patterns in the nutritional composition of grocery products. Several nutrients show strong relationships, particularly fat and calories. These insights provide a foundation for later steps in the project, such as clustering and dimensionality reduction.